

System Driven Hardware Design

- Constant False Alarm Rate Detection (CFAR) -

International Master of Science

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- Version 1,6 -



Automatically detect movements by a fixed probability of false alarm.

Until now:

- Performing DFT with windowing. You can see spectral peaks at low frequencies while movement is present.

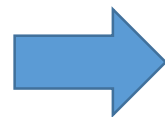
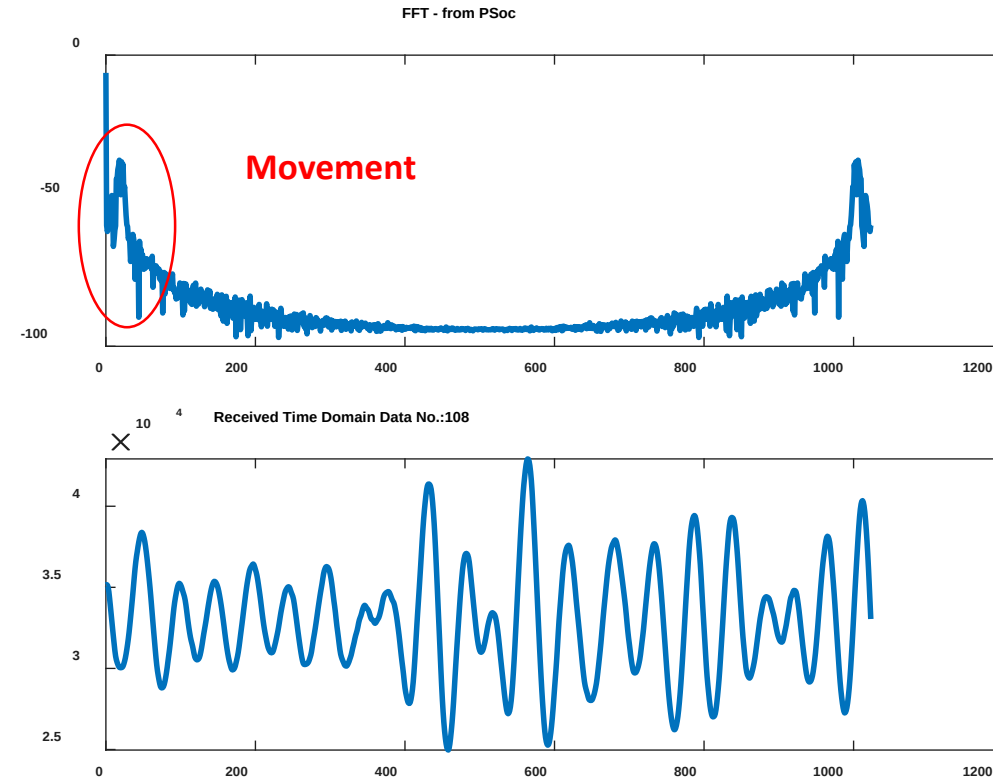
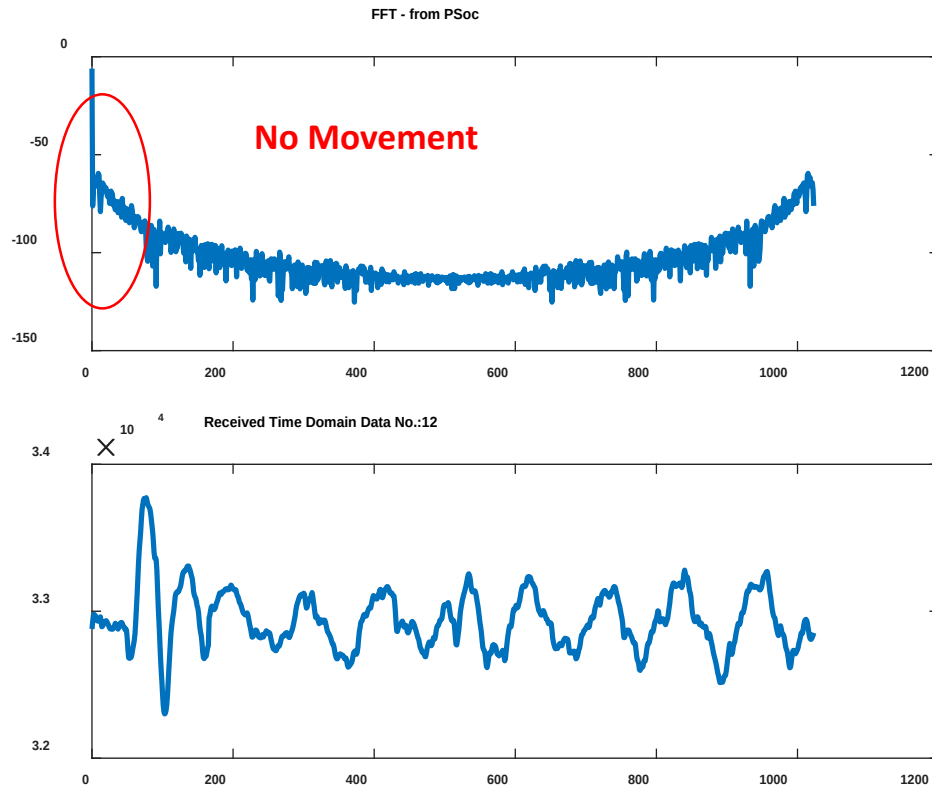
Now:

- How can we automatically detect movement and give alarm?
- How can we reduce the probability of false alarms?



We look at the CFAR Algorithm to solve this Questions.

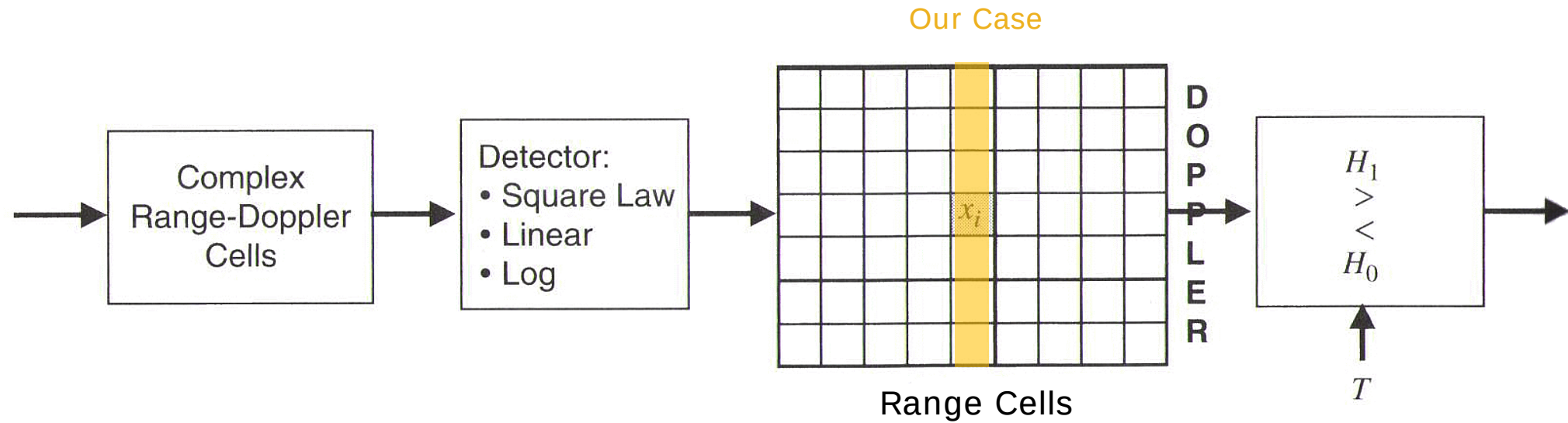
Motivation: Automatically detect movements in the presence of interferences



How to determine a threshold ?

Agenda

- Target Detection as Hypothesis Testing
- CFAR Theory
- CFAR in Matlab



T : threshold for testing x_i

Radar Echoes and interference are both described best with statistical methods

- *Interference* is modeled as zero mean additive white Gaussian (AWGN) noise (m_w and σ_w)
- *Target Signals* are modeled as constant present or not

Hypothesis Testing

H_0 : The measurement y is the result of interference only.

$\rightarrow p_y(y|H_0)$

H_1 : The measurement y is the result of interference and echoes from a target.

$\rightarrow p_y(y|H_1)$

} Probability
density
function
(pdf)

The following Probabilities are of interest

P_D : A target is declared when a target is in fact present $\rightarrow$ Detection.

P_M : A target is *not* declared when a target is in fact present $\rightarrow$ Missed.

P_{FA} : A target is declared when a target is in fact *not* present $\rightarrow$ False Alarm.

For the derivation in this section we assume: we do now the interference (noise) power $(\sigma_w)^2$, which is assumed to be constant.

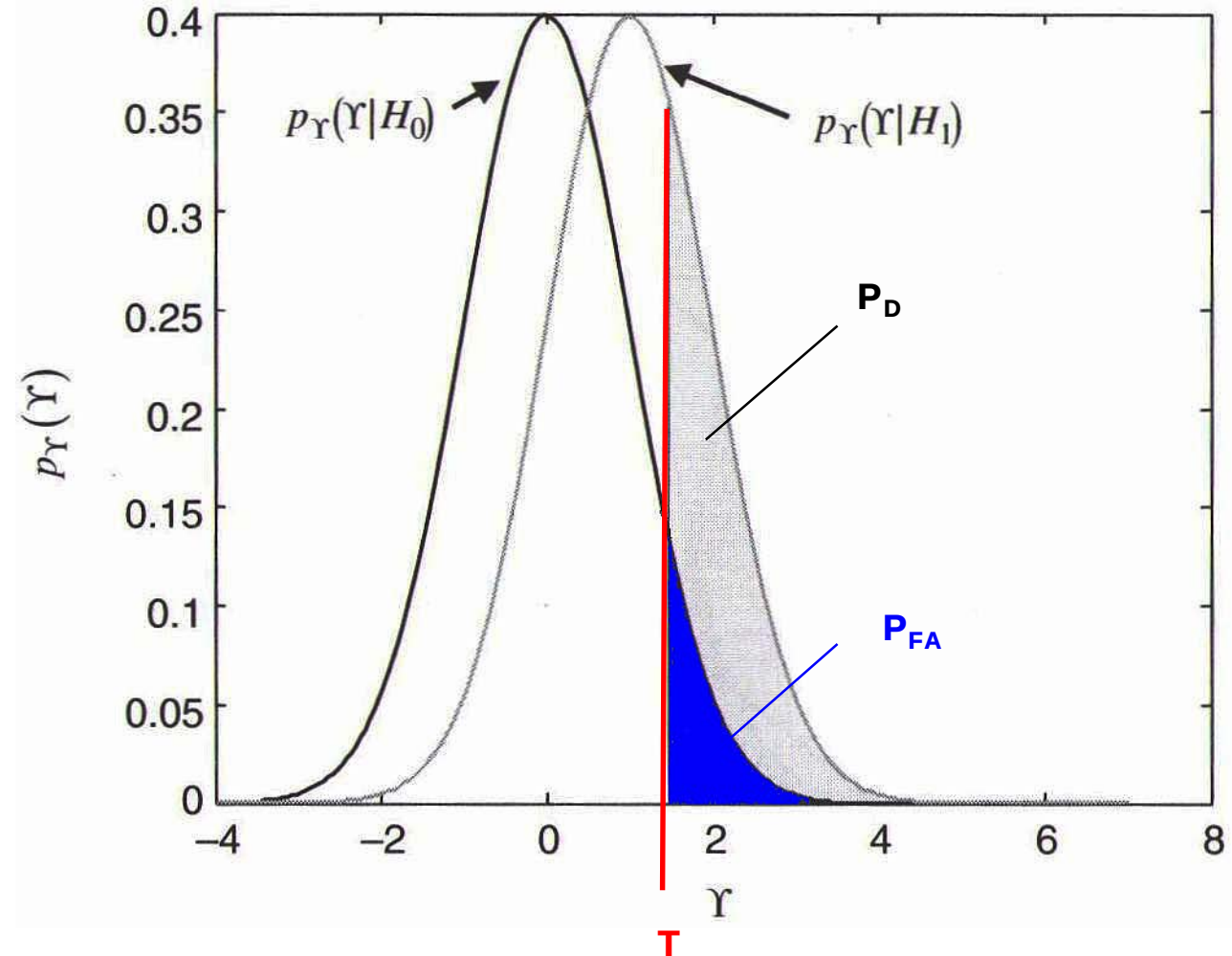
Likelihood Ratio

$$\frac{p(y|H_1)}{p(y|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} T$$

Decision rule

Choose T such that P_D is maximized,
subjected to $P_{FA} \leq \alpha$

Further Reading: [1] Chapter 6.1



1. Observation: P_D and P_{FA} grow or shrink together
2. Task: How to determine T ?

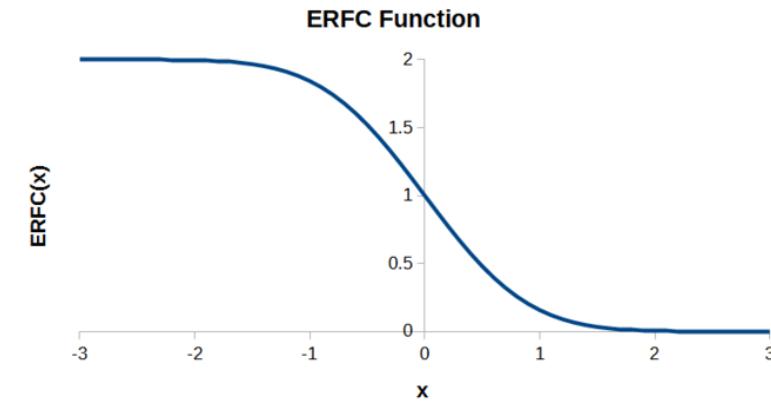
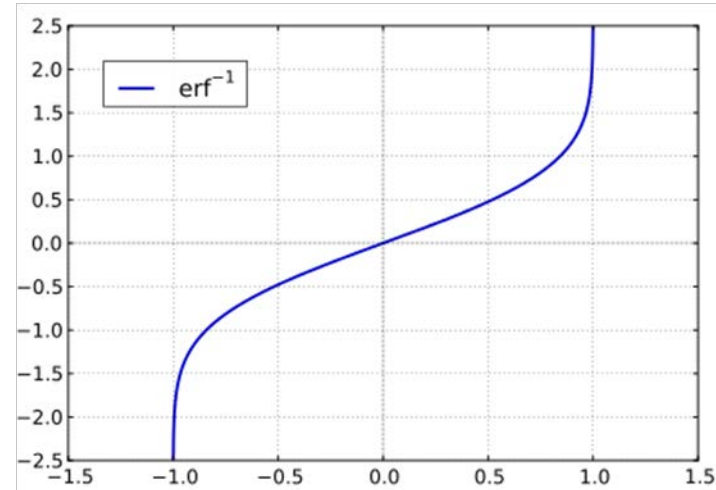
Resulting T and PD for a given P_{FA}

$$T = \sqrt{2N\sigma_w^2} \operatorname{erf}^{-1}(1 - 2P_{FA})$$

N: number of Samples

σ_w^2 : variance of the Interference (i.e. noise power)

m: magnitude of the target signal

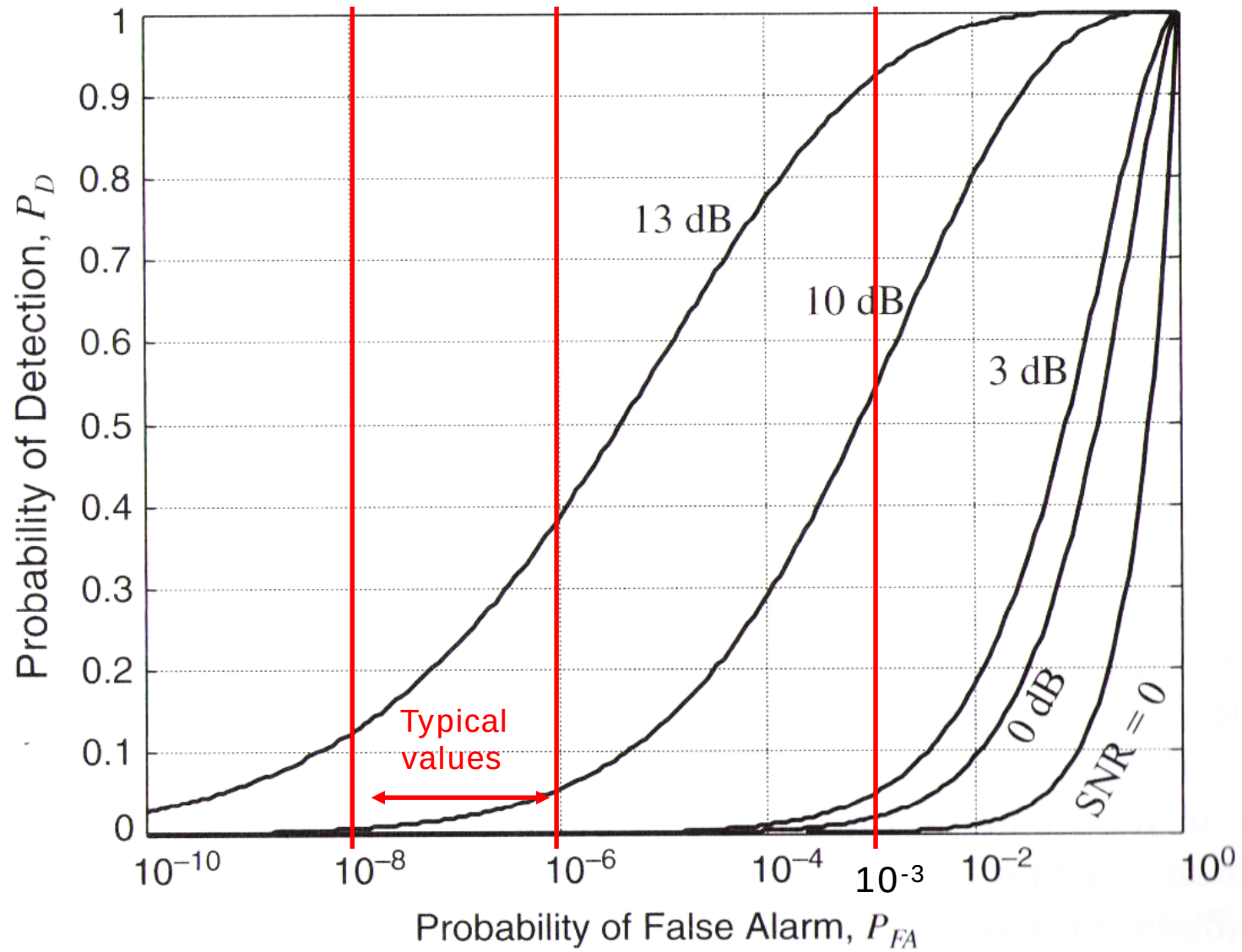


$$P_D = \frac{1}{2} \operatorname{erfc} \left(\operatorname{erfc}^{-1}(2P_{FA}) - \sqrt{\frac{Nm^2}{2\sigma_w^2}} \right)$$

$$P_D = \frac{1}{2} \operatorname{erfc} \left(\operatorname{erfc}^{-1}(2P_{FA}) - \sqrt{\frac{SNR}{2}} \right)$$



T depends on P_{FA} .
 P_D depends on SNR and P_{FA} .



Again look at

$$P_D = \frac{1}{2} \operatorname{erfc} \left(\operatorname{erfc}^{-1}(2 P_{FA}) - \sqrt{\frac{SNR}{2}} \right)$$

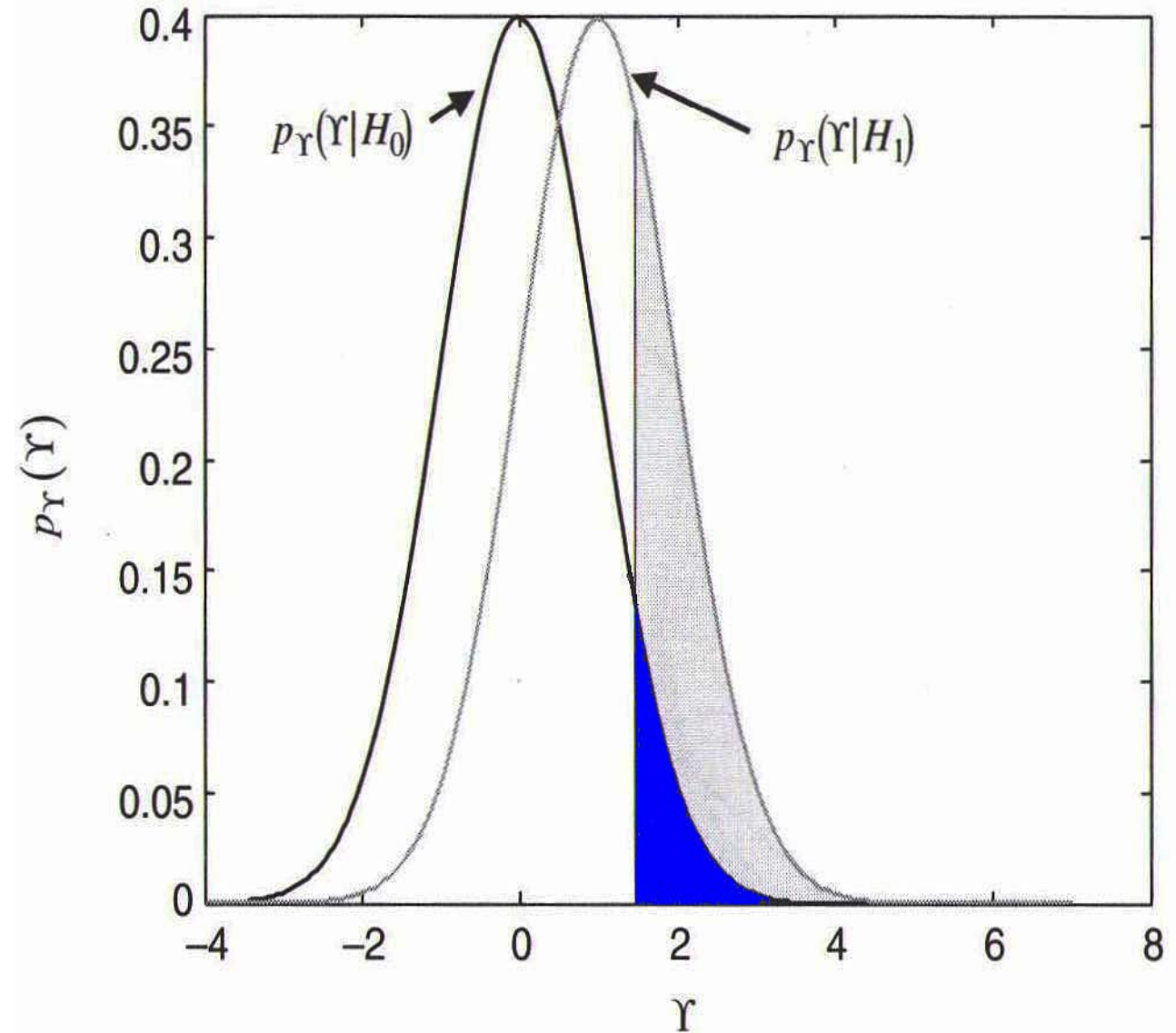
$$SNR = \frac{P_{sig}}{P_{noise}}$$



Increase the SNR



Recall: We assumed a **known** interference (noise) power $(\sigma_w)^2$



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- **CFAR Theory**
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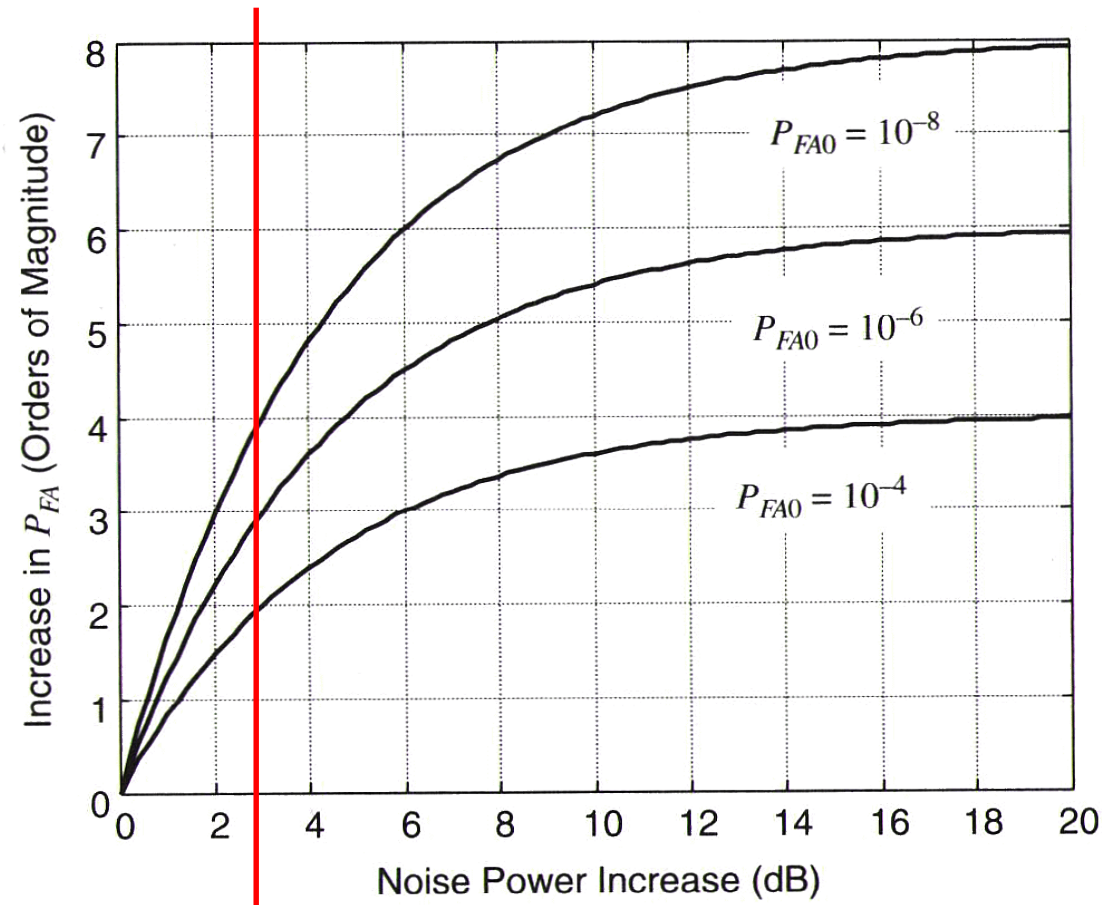
Now we are looking at a realistic situation: **unknown** interference (noise) power $(\sigma_w)^2$

Q: How to determine the threshold T?

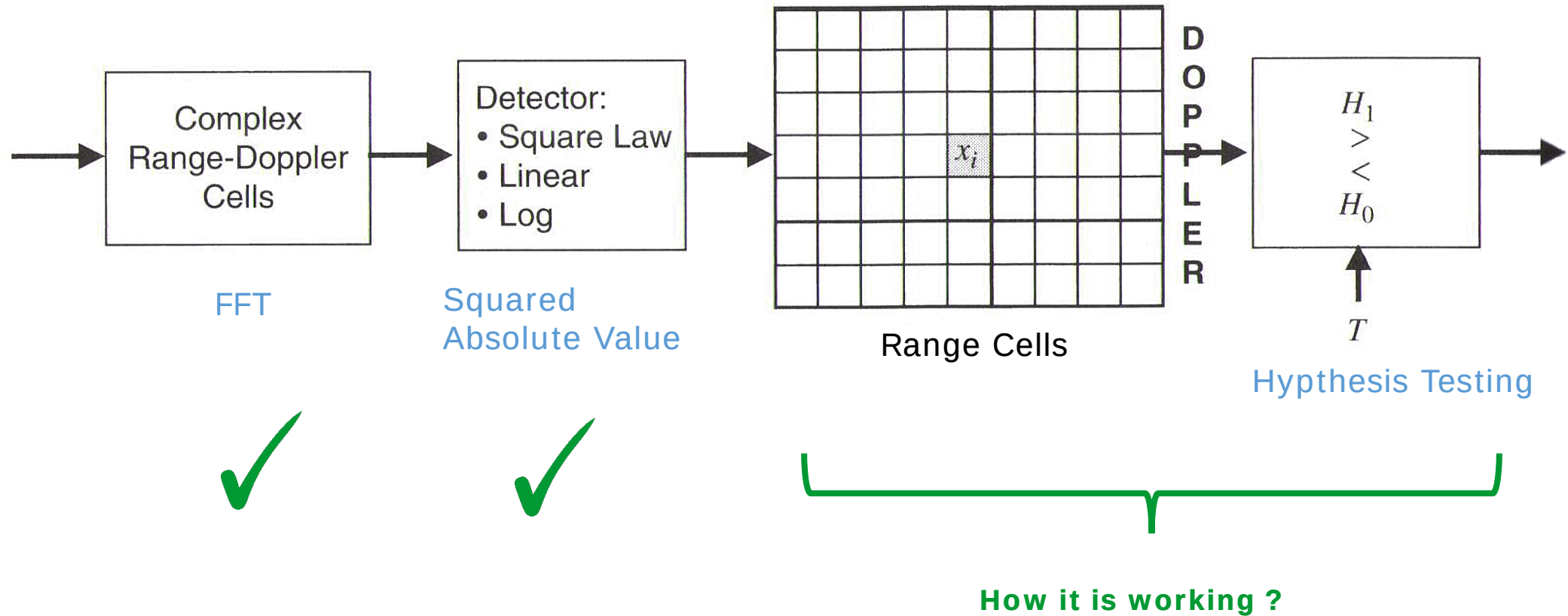
A: A Detection processor with constant P_{FA} and adaptive threshold T.

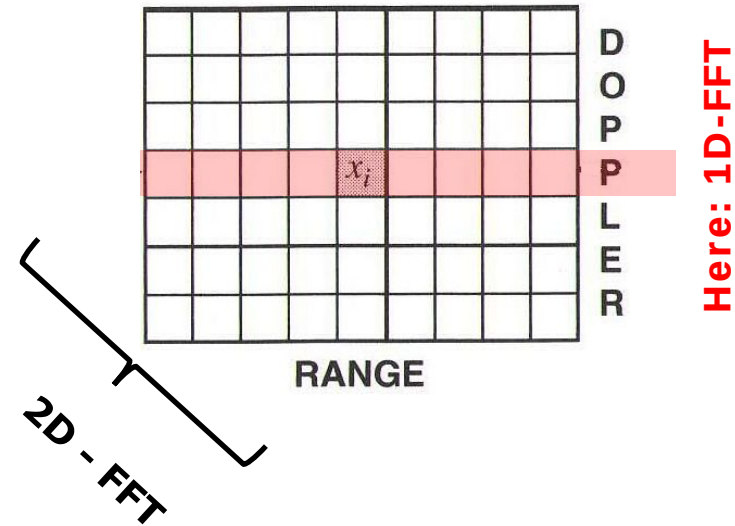
Note:

A Threshold T has been set for a known P_{FA0} . Then an Increase in Noise Power has been taken place for the previously calculated and fixed Threshold.



CFAR in order to make P_{FA} robust in changes of the interference level.
CFAR regarding a changing interference level.





Task

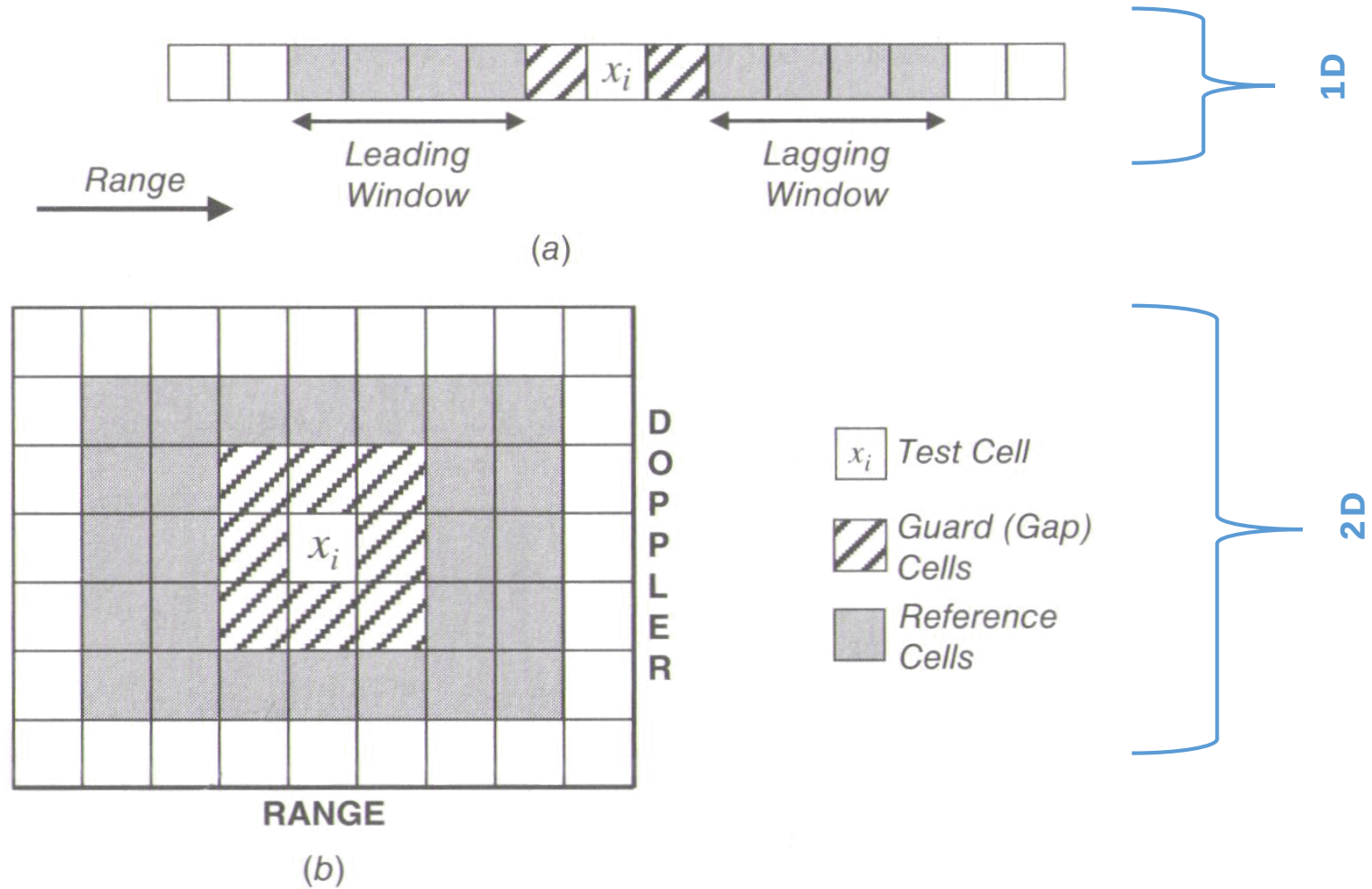
- Cell under test (CUT) is x_i
- Noise in the CUT determines the threshold
- The noise parameters in the CUT has to be estimated (here: σ_w^2)

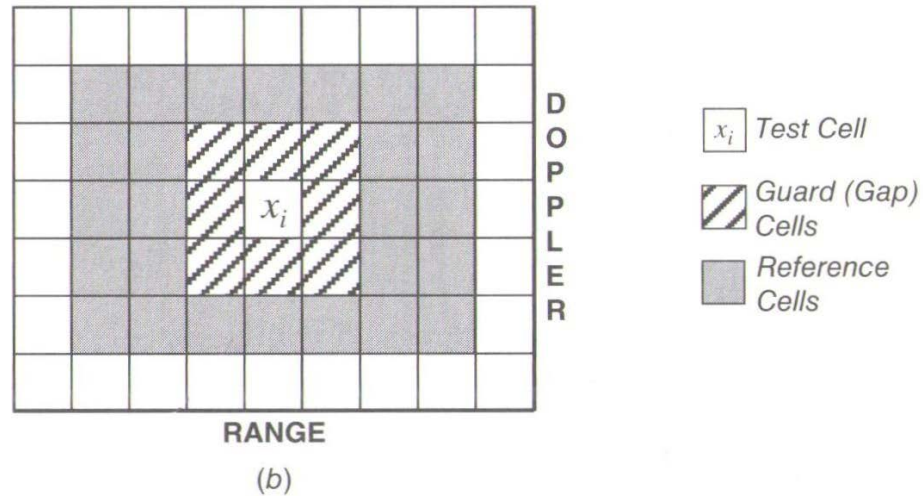
Assumptions

- The neighboring cells contain interference with the same statistics as the CUT (called *homogeneous* interference), so that they are representatives of the interference that is competing with the potential target.
- The neighboring cell do *not* contain any targets; they contain interference only



The interference statistics in the CUT can be estimated from the measured samples from the adjoining cells.





Noise estimate in the reference cells

$$\hat{\sigma}_w^2 = \frac{1}{N} \sum_{k=1}^N x_{ref}$$

$$x_{ref}(k) = |DFT(k)|^2$$

Required Threshold

$$\hat{T} = \alpha \hat{\sigma}_w^2$$

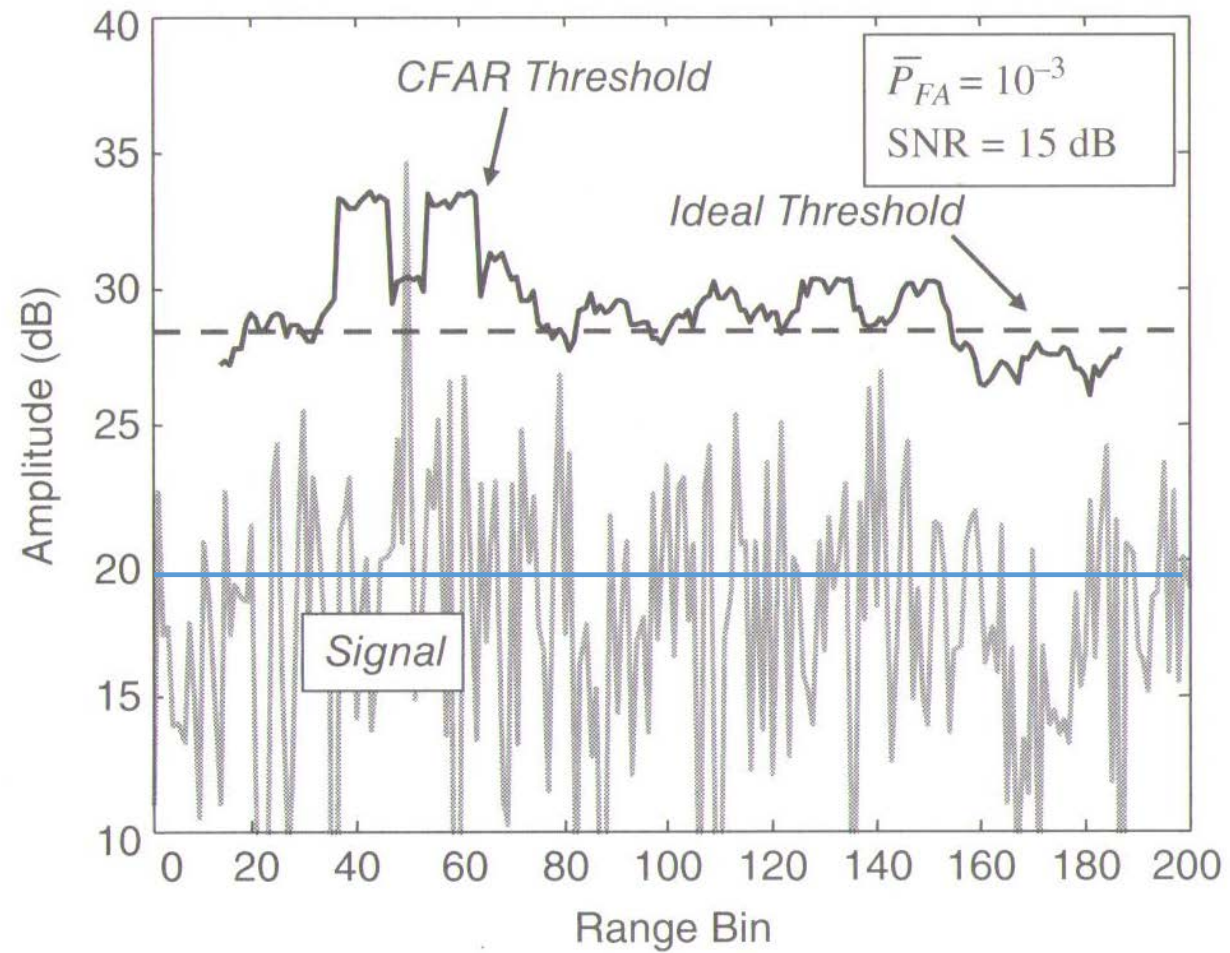
For α we obtain:

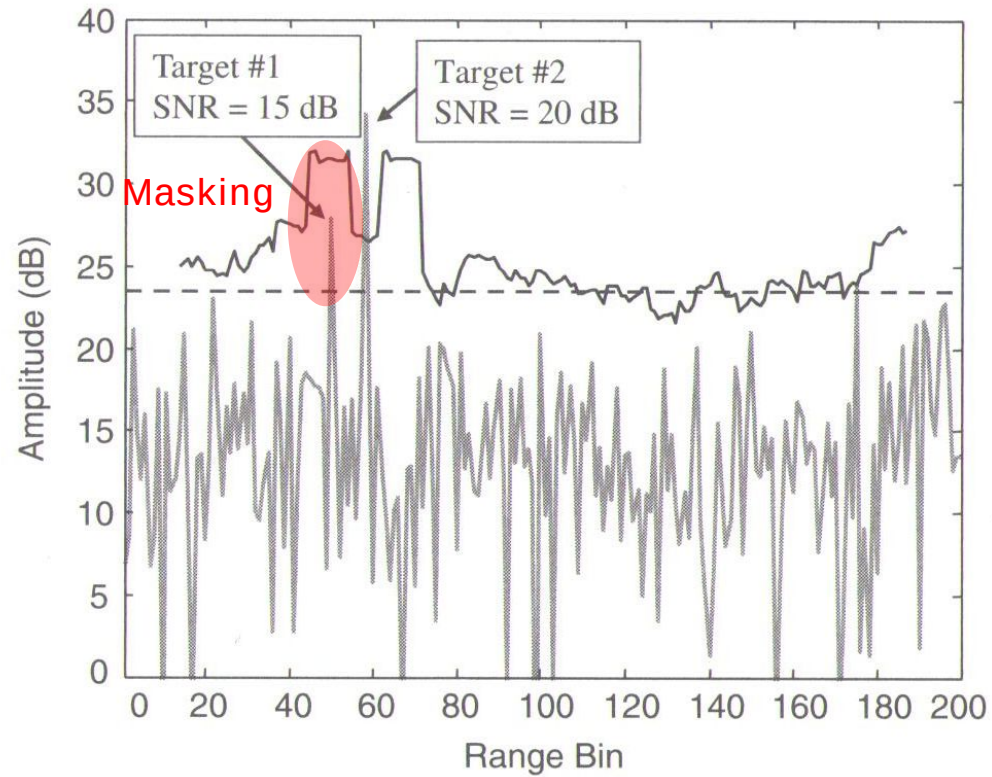
$$\alpha = N \left(\bar{P}_{FA}^{-1/N} - 1 \right)$$

The cell averaging techniques exhibits CFAR behavior.

with:

N number of averaged cells
P_{FA} target false alarm probability





Realistic Case

- Interference is caused by clutter
- No Targets in the reference cells

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Moodle: Download CA_CFAR_Student.m

Implement in Matlab

- Spectral Representation of the test signal
- Implement the CA_CFAR with the following parameters:
 - N: Number of Samples taken
 - NG: number of guard cells
 - NR: number of reference cells
 - PFA: Probability of false alarm
- Use the given ADC-Samples of a real measurement and test your CFAR Algorithm
- Investigate which N, NG and NR performs best on the ADC Data. Explain your choice
- In order to test your CFAR Algorithm:
 - Use the transferred ADC data in Matlab generated with your Radar Module

Implement on the PSoC Device

Transfer the CFAR Algorithm from Matlab To C

In order to test your C-code implementation: transfer both to Matlab: the ADC Data and the in C performed CFAR Algorithm

Turn on the red LED

If your CFAR in C is working properly: Turn off your UART transfer, run CFAR solely on your PSoC and turn on the RED LED for 0,5s, if you detect a movement.



Literature

1. *“Fundamentals of Radar Signal Processing”*, Mac Graw-Hill, Mark A. Richards, -> Section 6.5
Constant False Alarm Rate Detection